

MODULE 02

Python for Data Science

AI with Data Science - Professional Learning Series

ARSHITH LMS | Professional Edition

Python for Data Science

Why Python for Data Science?

Python is the #1 language for AI and Data Science because:

- Simple, readable syntax
- Massive ecosystem of libraries
- Strong community support
- Integration with ML frameworks
- Runs everywhere (cloud, local, notebooks)

Essential Libraries

```
import numpy as np # numerical computing
import pandas as pd # data manipulation
import matplotlib.pyplot as plt # visualization
import seaborn as sns # statistical plots
from sklearn import ... # machine learning
import tensorflow as tf # deep learning
import torch # deep learning
```

NumPy Basics

```
import numpy as np

# Create arrays
a = np.array([1, 2, 3, 4, 5])
b = np.zeros((3, 4)) # 3x4 zeros
c = np.ones((2, 3)) # 2x3 ones
d = np.random.rand(5, 5) # random 5x5

# Operations
print(a.shape, a.dtype, a.mean())
print(np.dot(b, c.T)) # matrix multiply
```

Pandas Basics

```
import pandas as pd

# Create DataFrame
df = pd.DataFrame({'name': ['Alice', 'Bob'], 'age': [25, 30], 'score': [85, 92]})

# Basic operations
print(df.head(), df.info(), df.describe())
df['grade'] = df['score'].apply(lambda x: 'A' if x >= 90 else 'B')
df_filtered = df[df['age'] > 25]
df_grouped = df.groupby('grade')['score'].mean()
```

Data Visualization

```
import matplotlib.pyplot as plt
import seaborn as sns

# Histogram
plt.hist(df['score'], bins=10, color='purple')
plt.title('Score Distribution')
plt.show()

# Scatter plot
plt.scatter(df['age'], df['score'], c='blue')

# Correlation heatmap
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.show()
```